↳ Suppose $S = \{w_1, w_2, \ldots, w_r\}$ is an orthogonal basis for a subspace W of a vector space V, then one may extended S to an orthogonal basis for V.

Positive definite matrices :-

Let A be a real symmetric matrix, then A is said to be positive definite if for every non-zero vector $u \in \mathbb{R}^n$ such that

$$\langle u, Au \rangle = (Au)^T u$$
$$= u^T A^T u$$
$$= u^T A u > 0$$

Ex:-

1) $A = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$, $|A| = -5 < 0$

$\therefore$ A is not positive definite

Properties of positive definite matrix :-

(i) $|A| > 0$

(ii) $\lambda_i > 0$

(iii) $tr(A) > 0$

(iv) determinant of all principal minors > 0.

(2) $\begin{bmatrix} 1 & -2 \\ -2 & 3 \end{bmatrix}$ → not P.d

(3) $A = \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix}$ $|A| = 1 > 0$

$|[a_{11}]| = 1 > 0$

$|[a_{22}]| = 5 > 0$

$\therefore$ A is positive definite matrix

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